Topic 8 – Features of manufacturing industries

8.1 Methods of production

Spec: Characteristics and stages of one-off, batch and high-volume production when applied to products and materials

Key facts

• **Characteristics of one-off production?** Unique/bespoke, skilled workforce, general-purpose tools, long time per item, high unit cost, high quality and flexibility.
• **Characteristics of batch production?** Set quantities, flexible general equipment, jigs/templates/moulds, semi-skilled to skilled workers, changeover downtime, moderate unit cost.
• **Characteristics of high-volume production?** Huge quantities, dedicated automated lines, expensive tooling, semi-skilled/unskilled labour, very low unit cost, inflexible.
• **Why batch produce 50 identical furniture items?** Jigs/templates reused to save marking-out time; general workshop equipment and skills can switch to other products; quantities match orders so little storage needed.
• **Why mass produce a bath toy?** Complex mould is expensive so high volumes needed to break even; large worldwide market; economies of scale reduce unit cost.
• **Stages in production planning?** Design for manufacture → choose materials/processes/scale → tooling and jigs → schedule → manufacture → QC → packaging and distribution.

	One-off (jobbing)	Batch	High-volume (mass)	Continuous
Quantity	1 (bespoke)	Specified quantity (tens–thousands), repeated as needed	Very large (thousands–millions)	Non-stop, 24/7
Workforce	Highly skilled craftspeople	Skilled/semi-skilled, flexible	Semi-skilled/unskilled + technicians, robots	Few operators, highly automated
Equipment	General-purpose hand and machine tools	General-purpose machines + jigs, templates, moulds; CNC	Dedicated automated lines, robots, expensive tooling	Dedicated plant
Flexibility	Very high	Medium – changeover between batches (downtime)	Low – retooling costly	Very low
Unit cost	Very high	Medium	Very low (economies of scale)	Lowest
Lead time	Long per item	Medium	Short per item, long set-up	Constant output
Examples	Speedboat, prototype, bespoke furniture, rocking chair, theatre set	Furniture ranges, clothing collections, bread, limited editions	Cars, phones, drinks bottles, toys	Steel, paper, chemicals, pipe extrusion, oil refining

Key terms

One-off production	Making a single, unique, often bespoke product to a client's specification – highly skilled, labour intensive, high unit cost (e.g. a speedboat built to order, a wedding dress, a prototype).
Batch production	Making a set quantity of identical products, then changing the set-up to make a different batch. Uses jigs, templates and moulds (e.g. 50 pieces of furniture, bakery goods).
High-volume (mass) production	Making very large numbers of identical products, often on automated production lines with semi-skilled labour and dedicated tooling (e.g. drinks bottles, cars).
Continuous production	24/7 non-stop production of a single product, often process industries (e.g. steel, paper, chemicals, extruded pipe).
Economies of scale	Reduction in unit cost as output increases, because fixed costs are spread across more products and materials are bought in bulk.
Unit cost	Total cost of making one product.
Downtime	Time when machines are not producing, e.g. during changeover between batches.

Exam tip (2022 Q6b, 2021 Q5b): Explain **why** the scale fits the context – link to tooling cost and break-even, market size, flexibility, storage, use of jigs and the skills of the workforce.

8.2 Quality monitoring systems

Spec: Quality control – the monitoring and achieving of high standards and degree of tolerance by inspection and testing, computer-aided testing; Quality assurance – monitoring the quality of a product from its design and development stage, through its manufacture, to its end-use performance and degree of customer satisfaction; Total Quality Management (TQM) – when applied to quality assurance procedures and its impact on employees at every stage of the production process, ISO 9000

Key facts

• **QC vs QA?** QC inspects and tests products to find faults; QA is the whole system of procedures that prevents faults from design to customer.
• **What does TQM involve?** Everyone responsible for quality, teamwork and quality circles, continuous improvement (kaizen), right first time, customer focus, training, empowerment.
• **Impact of TQM on employees?** More responsibility and involvement, training, teamwork, motivation and job satisfaction – but pressure and need for commitment.
• **Benefits of ISO 9001 certification for a car component maker?** Proves consistent quality to customers, needed to win contracts, fewer defects/returns, efficient documented processes, international recognition.
• **Disadvantages of QC systems?** Set-up and running costs, can slow production, sampling lets some faults through, destructive testing wastes products, equipment needs calibrating, diverts skilled staff.
• **Examples of computer-aided testing?** CMM probes, laser scanning against CAD, vision systems checking labels/colour/barcodes, X-ray/ultrasound for internal flaws, automatic weighing, sensors with feedback to adjust machines.
• **Name three types of quality monitoring system.** Quality control, quality assurance (ISO 9000), Total Quality Management (also Six Sigma).
• **Probability both QC tests fail if 98.5% and 98% pass?** $0.015 \times 0.02 = 0.0003$.

Key terms

Quality control (QC)	Inspection and testing of materials, components and products at stages during and after manufacture to check they meet standards and tolerances. Detects faults.
Quality assurance (QA)	A planned system that builds quality in at every stage – from design and suppliers through manufacture to end-use and customer satisfaction. Prevents faults.
Total Quality Management (TQM)	A management philosophy where every employee at every stage is responsible for quality and continuous improvement (kaizen), aiming for zero defects and customer satisfaction.
ISO 9000/9001	International standards for quality management systems; certification shows a company has documented, audited QA procedures.
Tolerance	The acceptable variation from a specified size, written as e.g. 50 ± 0.1 mm.
Destructive testing	Testing a sample until it fails (tensile, impact) – the sample is destroyed.
Non-destructive testing	Testing without damage – visual, X-ray, ultrasound, dye penetrant, CMM measurement.
Computer-aided testing	Automated inspection using sensors, cameras, CMMs and laser scanners linked to computers that compare results with CAD data.
CMM	Coordinate measuring machine – a probe (touch or laser) measures points on a part in 3D and compares with the CAD model.

Exam tip: For a one-off product (2023 Q6c) every boat is inspected – **stage testing** (laminate thickness, voids) and **end testing** (sea trials, leak tests) – while QA covers procedures, training, ISO 9001 and traceability.

8.3a Scheduling, robotics and materials handling

Spec: Production scheduling and production logistics; Robotics in production – robots on fully-automated production and assembly lines/cells; Materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)

Key facts

• **What does production scheduling aim to do?** Meet delivery dates, balance workloads, maximise machine use, minimise downtime and work-in-progress, coordinate labour and materials.

• **How does MPS software support JIT?** Triggers orders to suppliers, controls stock, tracks progress via barcodes/RFID, schedules machines and people, links to EPOS sales data.
• **Advantages of robots in production?** Work 24/7 without breaks, highly accurate and consistent, fast, work in hazardous environments (painting, welding), reduce labour costs, easily reprogrammed for new models.
• **Disadvantages of robots?** High initial cost, programming and maintenance skills, breakdown stops the line, cannot cope well with unexpected problems, job losses for unskilled workers.
• **How have computers made robots more effective?** Faster processing for precise motion, sensors and feedback (vision, force) for quality checks and self-diagnosis, integration with CAD/ERP/JIT, AI and reprogramming for different products.
• **Benefits of ASRS?** Saves floor space (tall, narrow aisles), fast and accurate picking (~99.9%), less labour, 24/7, integrates with stock control, safer.
• **How are AGVs guided?** Buried wires, magnetic/coloured tape, laser reflectors, cameras/vision, gyroscopes and floor transponders.
• **Name two automated materials handling systems.** ASRS and AGVs (also automated conveyors, robotic forklifts, automated pallet trucks).

Advantages	Disadvantages
Work 24/7 without breaks or fatigue	Very high capital cost and integration cost
High accuracy, repeatability and quality	Need skilled programmers and maintenance engineers
Fast – higher productivity, lower unit cost	Breakdowns can halt a whole line
Do dangerous, dirty, repetitive jobs (painting, welding, heavy lifting)	Limited ability to handle unexpected problems
Reprogrammable for new products/models	Job losses for semi-skilled workers

Key terms

Production scheduling	Planning what will be made, when, on which machines and by whom, so orders are completed on time with efficient use of resources.
Production logistics	Organising the flow of materials, components, information and finished goods through the supply chain – from suppliers, through the factory, to customers.
Master production schedule (MPS)	Software plan stating which products will be made in each period; links to stock control, ordering and JIT.
Gantt chart	A bar chart showing tasks against time – used to plan and monitor schedules.
Industrial robot	Programmable, multi-axis (often 6-axis) manipulator that performs tasks such as welding, painting, assembly, pick-and-place and palletising.
End effector	The tool on the end of a robot arm – gripper, welding torch, spray gun, vacuum cup.
Cobot	Collaborative robot designed to work safely alongside people.
ASRS	Automated storage and retrieval system – computer-controlled cranes/shuttles in tall racking that store and fetch items automatically.
AGV	Automatic guided vehicle – driverless vehicle that moves materials around a factory, guided by wires, magnetic tape, lasers or cameras.

8.3b FMS, lean manufacturing and standardised parts

Spec: Flexible manufacturing systems (FMS), modular/cell production systems; Lean manufacturing using just-in-time (JIT) systems; Standardised parts, bought-in components

Key facts

• **Two main features of FMS?** Modular/cell production (CNC machines + robots + AGVs in cells) and SMED (quick tool/die changes).
• **Advantages of FMS?** Quick changeover between products, small batches economically, shorter lead times, high quality, lower labour costs, supports JIT and mass customisation.
• **Disadvantages of FMS?** Very high set-up cost, complex planning and programming, skilled maintenance, can be less efficient than a dedicated line for one product.
• **Five principles of lean manufacturing?** Value, value stream, flow, pull, perfection.
• **The 'seven wastes' (TIMWOOD)?** Transport, Inventory, Motion, Waiting, Over-production, Over-processing, Defects.
• **Benefits of JIT?** Less stock and storage space, less capital tied up, better cash flow, less waste/damage/obsolescence, materials arrive in order of use, responds to customer orders.
• **Risks of JIT?** Total reliance on suppliers – strikes, transport problems, pandemics or extreme weather stop production; needs excellent logistics and quality.
• **Benefits of standardised parts and bought-in components?** Cheaper (bulk/suppliers' economies of scale), already tested and reliable, reduce development time, interchangeable, easy replacement parts, frees in-house staff, less tooling.
• **Drawbacks of bought-in components?** Less control over quality and supply, reliance on suppliers, less unique product, may not be ideal fit.

Advantages	Disadvantages
Rapid response to demand, small batches, mass customisation	Very expensive to set up
Shorter lead times for new products	Complex planning, programming and scheduling
High, consistent quality; less labour	Needs highly skilled staff and maintenance
Supports JIT and QRM; teamwork in cells improves motivation	Less efficient than a dedicated line for one huge-volume product

Key terms

Flexible manufacturing system (FMS)	A computer-controlled group of CNC machines, robots and AGVs that can quickly switch between making different products in response to demand.
Cell (modular) production	Grouping machines and multi-skilled workers into self-contained cells that make a complete product or product family.
SMED	Single Minute Exchange of Dies – techniques to change tooling in under 10 minutes, reducing downtime.
Lean manufacturing	A philosophy (from Toyota) of eliminating waste and adding value at every stage, using five principles: value, value stream, flow, pull, perfection.
Just-in-time (JIT)	Materials and components arrive exactly when needed for production – no large stocks are held.
Kanban	A visual signal (card, bin, electronic) that triggers the production or delivery of more parts when they are used – a pull system.
Kaizen	Continuous improvement through small, regular changes involving all employees.
Standardised parts	Common components made to agreed standard sizes (e.g. M6 bolts, AA batteries, bearings) used across many products.
Bought-in components	Parts bought from specialist suppliers rather than made in-house (e.g. motors, switches, chips, castors).

8.3c QRM, data integration and concurrent manufacturing

Spec: Quick response manufacturing (QRM); Data integration – product data management (PDM), enterprise resource planning (ERP) systems; Concurrent manufacturing

Key facts

• **What is the main aim of QRM?** Reduce lead times across the whole company so customised products reach customers quickly.
• **Features of QRM?** Flow (not batch) production triggered by demand, flexible manufacturing cells, multi-skilled teams owning product families, JIT, TQM/Six Sigma, rapid prototyping, POLCA cards.
• **Benefits of QRM to a manufacturer?** Shorter lead times and time to market, respond quickly to trends, customisation, lower stock/storage/working capital, improved cash flow, less waste, better customer satisfaction and market share.
• **Drawbacks of QRM?** Reorganisation and training costs, reliance on supply chain, may struggle with sudden large volume increases, changes employees' roles.
• **Example of a QRM company?** Zara – new fashion designs in stores within weeks; also specialist vehicle builders.
• **What data does PDM hold?** CAD models and drawings, specifications, materials, BOM, manufacturing details, costs, suppliers, revision history, sign-offs.
• **Benefits of ERP?** One shared real-time database across departments, accurate stock and planning, supports JIT, better decisions, removes duplicated data entry, improves customer service.
• **Disadvantages of ERP?** Very expensive and complex to implement, staff training, customisation, dependence on IT, data security risks.

• **Concurrent vs linear (sequential) manufacturing?** Linear: each department finishes before the next starts – slow, late changes costly. Concurrent: tasks overlap with multidisciplinary teams – faster, right first time.
• **Benefits of concurrent manufacturing?** Shorter development time and time to market, fewer late design changes, lower costs, better quality, improved communication and teamwork.

Benefits (2023 Q9)	Drawbacks
Shorter lead times and time to market	Major reorganisation and training costs
Rapid response to trends and changing designs	Heavy reliance on reliable suppliers – strikes/transport delays stop work
Customisation and flexibility	May struggle with very large, sudden increases in demand
Less stock and work-in-progress; lower storage costs	Changes employees' roles and responsibilities
Lower working capital, improved cash flow	
Less waste, teams own quality, better job satisfaction	
Customer satisfaction, reputation and market share	

Key terms

Quick response manufacturing (QRM)	A company-wide strategy to reduce lead times in all areas (office and shop floor), moving from batch to flow production triggered by customer demand, using flexible cells and multi-skilled teams. Suited to low-volume, customised products.
Lead time	The time from receiving an order to delivering the product.
POLCA	Paired-cell Overlapping Loops of Cards with Authorisation – the card system used in QRM to control flow between cells.
Product data management (PDM)	Software that stores and controls all product data – CAD models, drawings, specifications, bills of materials, revisions – in one secure place so everyone uses the latest version.
Bill of materials (BOM)	A list of every part, sub-assembly and material, with quantities, needed to make a product.
Enterprise resource planning (ERP)	Integrated software linking all business functions – sales, purchasing, stock, production planning, finance, HR and logistics – using one shared database.
Concurrent manufacturing (engineering)	Carrying out design, development, manufacturing planning, sourcing and marketing tasks in parallel by multidisciplinary teams, rather than one after another.
Time to market	The time taken from product concept to the product being on sale.